

Why a Shortest Elastic-Reduction Path Contains No Integer Rotation

Technical note

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Abstract

The unit-shear elastic reduction searches words in four integer simple shears. This note proves that a globally shortest word reaching the elastic domain cannot contain a nonempty path segment whose product is a lattice rotation. The proof uses two exact facts: the elastic domain is invariant under integer rotations, and conjugating a unit shear by an integer rotation gives another unit shear. A rotational segment can therefore be moved to the end of the path and removed, contradicting shortestness. This makes the comparison between the two elastic quadrants precise. The shortest path is rotation-free in the path-segment sense, while, for a generic elastic representative, every path to the other quadrant differs from it by a quarter turn. A longer word need not display that quarter turn as a visible three-shear block; its rotational content is instead exposed by the relative matrix quotient.

1 Definitions and the exact claim

Let the four allowed unit-shear generators be

$$\mathcal{S} = \{U_+, U_-, L_+, L_-\}, \quad U_+ = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad L_+ = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad U_- = U_+^{-1}, \quad L_- = L_+^{-1}. \quad (1)$$

A BFS path is a word $w = s_1 s_2 \cdots s_n$ with $s_i \in \mathcal{S}$, and $\Pi(w) = s_1 s_2 \cdots s_n \in \mathrm{SL}(2, \mathbb{Z})$ is its accumulated right-side transformation. Starting from a positive-definite metric $\mathbf{C}_0$, the endpoint is

$$\mathbf{C}_w = \Pi(w)^\top \mathbf{C}_0 \Pi(w). \quad (2)$$

The integer proper-rotation group is

$$\mathcal{K} = \mathrm{SO}(2) \cap \mathrm{SL}(2, \mathbb{Z}) = \{\mathbf{I}, \mathbf{Q}, -\mathbf{I}, -\mathbf{Q}\}, \quad \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (3)$$

These are rotations through $0, -\pi/2, \pi, \pi/2$, respectively. There are no others: every column of an integer orthogonal matrix is an integer unit vector, hence one of the signed coordinate vectors.

Definition 1 (Rotational segment). A word w *contains a nontrivial rotation* if it has a nonempty contiguous subword $\rho = s_i \cdots s_j$ for which $\Pi(\rho) \in \mathcal{K} \setminus \{\mathbf{I}\}$.

Contiguity is essential: a path segment consists of consecutive moves. It would not be meaningful merely to ask whether the final matrix admits some factorization containing $\mathbf{Q}$, because every matrix can be written trivially as $\mathbf{M} = (\mathbf{M}\mathbf{Q}^{-1})\mathbf{Q}$.

The theoretical elastic domain tested by the implementation can be written compactly as

$$\mathcal{E} = \left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix} \succ 0 : 2|b| \leq \min(a, c) \right\}. \quad (4)$$

Indeed, the four sign-and-swap variants of the fundamental inequalities $b \geq 0$, $c \geq a$, and $2b \leq a$ give exactly Equation (4). A *globally shortest elastic word* is a word of minimum length among all words satisfying $\mathbf{C}_w \in \mathcal{E}$. This is the notion of “short path” used below. It minimizes over the full elastic domain, not over a preselected quadrant.

2 The two algebraic ingredients

First, the elastic domain is invariant under every $\mathbf{R} \in \mathcal{K}$. It is enough to check the generator $\mathbf{Q}$:

$$\mathbf{Q}^\top \begin{pmatrix} a & b \\ b & c \end{pmatrix} \mathbf{Q} = \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}. \quad (5)$$

This swaps a and c and changes the sign of b , leaving $2|b| \leq \min(a, c)$ unchanged. Therefore

$$\mathbf{C} \in \mathcal{E} \iff \mathbf{R}^\top \mathbf{C} \mathbf{R} \in \mathcal{E} \quad (\mathbf{R} \in \mathcal{K}). \quad (6)$$

Second, $\mathcal{K}$ normalizes the unit-shear generator set. Direct multiplication gives

$$\begin{aligned} \mathbf{Q}U_+\mathbf{Q}^{-1} &= L_-, & \mathbf{Q}U_-\mathbf{Q}^{-1} &= L_+, \\ \mathbf{Q}L_+\mathbf{Q}^{-1} &= U_-, & \mathbf{Q}L_-\mathbf{Q}^{-1} &= U_+. \end{aligned} \quad (7)$$

Conjugation by $-\mathbf{I}$ does nothing, and conjugation by $-\mathbf{Q}$ is the same as conjugation by $\mathbf{Q}$. Hence

$$\mathbf{R}\mathcal{S}\mathbf{R}^{-1} = \mathcal{S} \quad (\mathbf{R} \in \mathcal{K}). \quad (8)$$

Consequently, if $v = t_1 \cdots t_m$ is a shear word, then $\mathbf{R}\Pi(v)\mathbf{R}^{-1}$ is represented by the word obtained by conjugating every t_i . The new word has exactly the same length.

3 The shortest path is rotation-free

Theorem 1 (No rotational segment in a shortest elastic word). *Let $\mathbf{C}_0 \succ 0$, and let w be a globally shortest word in $\mathcal{S}$ for which $\mathbf{C}_w \in \mathcal{E}$. Then w contains no nonempty contiguous subword whose product belongs to $\mathcal{K} \setminus \{\mathbf{I}\}$.*

Proof. Assume the opposite. Factor the word as

$$w = \alpha \rho \beta, \quad \Pi(\alpha) = \mathbf{A}, \quad \Pi(\rho) = \mathbf{R} \in \mathcal{K} \setminus \{\mathbf{I}\}, \quad \Pi(\beta) = \mathbf{B}. \quad (9)$$

Because $\mathbf{R}$ normalizes $\mathcal{S}$, conjugating every letter of β produces another shear word β' of the same length, with

$$\Pi(\beta') = \mathbf{R}\mathbf{B}\mathbf{R}^{-1}. \quad (10)$$

The total matrix can now be rearranged exactly:

$$\Pi(w) = \mathbf{A}\mathbf{R}\mathbf{B} = \mathbf{A}(\mathbf{R}\mathbf{B}\mathbf{R}^{-1})\mathbf{R} = \underbrace{\mathbf{A}\Pi(\beta')}_{\widetilde{\mathbf{M}}} \mathbf{R}. \quad (11)$$

Thus the original elastic endpoint is

$$\mathbf{C}_w = \mathbf{R}^\top (\widetilde{\mathbf{M}}^\top \mathbf{C}_0 \widetilde{\mathbf{M}}) \mathbf{R} \in \mathcal{E}. \quad (12)$$

By rotation-invariance of $\mathcal{E}$, it follows that

$$\widetilde{\mathbf{M}}^\top \mathbf{C}_0 \widetilde{\mathbf{M}} \in \mathcal{E}. \quad (13)$$

But $\widetilde{\mathbf{M}}$ is represented by the word $\alpha\beta'$, whose length is

$$|\alpha\beta'| = |w| - |\rho| < |w|. \quad (14)$$

This is a strictly shorter path into the elastic domain, contradicting the minimality of w . If joining α to β' creates adjacent inverse shears, cancelling them only shortens the replacement further. $\square$

Corollary 1. *A globally shortest elastic word also contains no nonempty contiguous subword with product $\mathbf{I}$.*

Proof. Such a subword can be deleted directly without changing the accumulated matrix. The resulting word reaches the same endpoint and is shorter. $\square$

What has been proved. The short path cannot hide a quarter turn, a half turn, or an identity loop as any consecutive sequence of its shear moves. The result is exact and does not require a generic metric, a unique endpoint, or a trivial lattice stabilizer. It relies only on Equations (6) and (8).

4 What this says about the other quadrant

Theorem 1 concerns the literal shear word. The complementary statement about *all* paths to the two elastic representatives is most cleanly expressed by a relative matrix quotient.

Let $\mathbf{M}_s$ be a shortest transformation and set

$$\mathbf{C}_s = \mathbf{M}_s^\top \mathbf{C}_0 \mathbf{M}_s. \quad (15)$$

For any other terminal transformation $\mathbf{M}$, define $\mathbf{R} = \mathbf{M}_s^{-1} \mathbf{M}$. Then

$$\mathbf{M}^\top \mathbf{C}_0 \mathbf{M} = \mathbf{R}^\top \mathbf{C}_s \mathbf{R}. \quad (16)$$

If the endpoint equals $\mathbf{C}_s$, then

$$\mathbf{R} \in \text{Aut}_{\mathbb{Z}}(\mathbf{C}_s) := \{\mathbf{A} \in \text{SL}(2, \mathbb{Z}) : \mathbf{A}^\top \mathbf{C}_s \mathbf{A} = \mathbf{C}_s\}. \quad (17)$$

If instead the endpoint is the quarter-turned representative $\mathbf{Q}^\top \mathbf{C}_s \mathbf{Q}$, then

$$\mathbf{R} = \mathbf{A} \mathbf{Q} \quad \text{for some} \quad \mathbf{A} \in \text{Aut}_{\mathbb{Z}}(\mathbf{C}_s). \quad (18)$$

For a generic positive-definite metric the integer stabilizer is only $\{\pm \mathbf{I}\}$. Equations (17)–(18) then reduce to the sharp classification

same representative: $\mathbf{M}_s^{-1} \mathbf{M} \in \{\pm \mathbf{I}\},$ other representative: $\mathbf{M}_s^{-1} \mathbf{M} \in \{\pm \mathbf{Q}\}. $
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(19)

This statement includes arbitrarily long words: however many extra shears a path uses, its net difference from the short path lies in the indicated rotation class whenever it reaches the same exact representative.

There is an important limitation. Equation (19) does *not* imply that every displayed longer word contains a contiguous three-letter copy of $\mathbf{Q}$. Group relations and cancellations can spread that rotational content throughout the word. What is true is:

1. the globally shortest word has no removable rotational segment, by Theorem 1; and
2. every word reaching the other generic representative has a net quarter-turn relative to that shortest word, by Equation (19).

Together these two statements justify assigning the extra lattice rotation to the other-quadrant description without falsely claiming that it must be visible as a local block in every possible word.

5 The simple-shear example

For

$$\mathbf{F} = \begin{pmatrix} 1 & 1.6 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{C}_0 = \mathbf{F}^\top \mathbf{F} = \begin{pmatrix} 1 & 1.6 \\ 1.6 & 3.56 \end{pmatrix}, \quad (20)$$

the shortest BFS word and the first word reaching the other representative are

$$w_s = U_- U_-, \quad \mathbf{M}_s = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad (21)$$

$$w_\ell = U_- L_- U_+, \quad \mathbf{M}_\ell = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}. \quad (22)$$

The short word plainly has no rotational subword; Theorem 1 shows that this is structural rather than an accident of its length. The relative quotient exposes the other path's quarter turn:

$$\mathbf{M}_s^{-1} \mathbf{M}_\ell = \mathbf{Q}, \quad \mathbf{M}_\ell = \mathbf{M}_s \mathbf{Q}. \quad (23)$$

At word level, the same calculation is

$$(U_- U_-)^{-1} (U_- L_- U_+) = U_+ U_+ U_- L_- U_+ \longrightarrow U_+ L_- U_+ = \mathbf{Q}, \quad (24)$$

where the arrow cancels the adjacent $U_+ U_-$ pair. The terminal metrics are therefore the two quarter-turned elastic representatives

$$\mathbf{C}_s = \begin{pmatrix} 1 & -0.4 \\ -0.4 & 1.16 \end{pmatrix}, \quad \mathbf{C}_\ell = \begin{pmatrix} 1.16 & 0.4 \\ 0.4 & 1 \end{pmatrix} = \mathbf{Q}^\top \mathbf{C}_s \mathbf{Q}. \quad (25)$$

Table 1: Computational checks used while developing the algebraic statement. They support the theorem but are not part of its proof.

Check	Cases examined	Failures
Global shortest word had a contiguous nontrivial rotational segment	73 generic starts	0
All terminal paths obeyed the same/other relative-rotation classification	2414 terminal paths from 75 starts	0

6 Conclusion

The missing logical step is now supplied: the shortest elastic-reduction path is not merely shorter than a path carrying a relative quarter turn; it is provably free of every removable integer-rotation segment. If such a segment existed, rotation invariance of the target and rotational closure of the shear generators would construct a strictly shorter elastic path.

For generic metrics, any path to the other elastic representative differs from this rotation-free shortest path by $\pm \mathbf{Q}$, while paths to the same representative differ by $\pm \mathbf{I}$. At exceptional symmetric metrics the stabilizer in Equation (17) can be larger, so the cosets in Equation (19) must be replaced by the full stabilizer and its $\mathbf{Q}$ -coset. The rotation-free theorem itself is unaffected by that exception.